



Aishu R 251001007-2025-IT ▾

A2**Started on** Monday, 24 August 2026, 12:37 PM**State** Finished**Completed on** Monday, 24 August 2026, 12:37 PM**Time taken** 33 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $arr[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  void sort(int a[], int n)
4  {
5      int i, j, temp;
6
7      for (i = 0; i < n - 1; i++)
8      {
9          for (j = 0; j < n - i - 1; j++)
10         {
11             if (a[j] > a[j + 1])
12             {
13                 temp = a[j];
14                 a[j] = a[j + 1];
15                 a[j + 1] = temp;
16             }
17         }
18     }
19 }
20
21 int main()
22 {
23     int n, i;
24     int a[1000];
25     long long sum = 0;
26
27     scanf("%d", &n);
28
29     for (i = 0; i < n; i++)
30         scanf("%d", &a[i]);
31
32     sort(a, n);
33
34     for (i = 0; i < n; i++)
35         sum += (long long)a[i] * i;
36
37     printf("%lld", sum);
38
39     return 0;
40 }
41

```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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